

RNA-seq and WSI-RNA Fusion Molecular Subtype Classification of TCGA-BRCA

1 Problem Definition

We evaluate a unimodal RNA-seq branch for **PAM50 molecular subtype classification** in the TCGA-BRCA cohort. This branch is intended as a molecular baseline and future fusion component for the existing histology branch.

The target consists of four PAM50 molecular subtypes:

Table 1: PAM50 classes used for the RNA-seq experiment.

Class	Label	Description
0	Luminal A	Hormone receptor-positive, low proliferation
1	Luminal B	Hormone receptor-positive, higher proliferation
2	Basal-like	Triple-negative, aggressive
3	HER2-enriched	HER2 overexpression

Samples labelled as Normal-like were excluded to match the current four-class WSI setup. The RNA-seq table contains 1,045 primary tumour samples. Each fold uses 837 training samples, 104 validation samples, and 104 test samples.

2 Pipeline

The classification pipeline first evaluates RNA-seq as a molecular branch, then reuses the same processed RNA table for WSI-RNA fusion.

2.1 Stage 1: RNA-seq Preprocessing

RNA-seq data were downloaded, filtered to the target PAM50 classes, converted into CLAM-compatible sample tables, and aligned with the patient-level fold splits. For the selected configuration, gene features are selected within each training fold by variance and standardised using training-fold statistics only.

2.2 Stage 2: RNA-level Classification – MLP

Sample-level predictions are made using a fully connected multilayer perceptron. The best configuration selected by the sweep uses:

Table 2: Best RNA configuration selected by Bayesian sweep on fold 0 validation AUC $\uparrow$.

Parameter	Value
Architecture	MLP
Hidden dimensions	[1024, 512]
Input genes	10000 top-variance genes
Dropout	0.4
Optimizer	AdamW
Learning rate	0.000197465
Weight decay	0.000700548
Label smoothing	0.05
Weighted loss	Enabled
Weighted sampling	Enabled
Batch size	64
Max epochs	80
Early stopping patience	16

2.3 Stage 3: WSI-RNA Fusion

The multimodal branch extends the existing CLAM pipeline. Each WSI bag is paired with RNA-seq features by TCGA case identifier. Slides without matched RNA data are excluded, WSI and RNA labels are checked for consistency, and the same patient-level folds are reused.

Within each fold, the RNA transform is fitted on the training split only and then applied to validation and test samples. The RNA vector is encoded by a two-layer MLP with hidden dimension 256. The WSI branch is CLAM-MB over UNI2-h slide features. Four fusion strategies are evaluated:

- **Concat fusion:** the CLAM slide embedding and RNA embedding are concatenated and passed through a fusion head with hidden dimension 32.
- **Gated fusion:** WSI and RNA embeddings are projected to 32 dimensions, a sigmoid gate is learned from the paired embeddings, and the final representation is the gated mixture of both modalities.
- **Late probability fusion:** RNA and WSI probabilities are linearly combined, with the RNA weight selected on the validation split.
- **Residual fusion:** a frozen RNA model provides the base logits, while a small WSI-conditioned head learns a conservative residual correction.

3 Experimental Setup

3.1 Hyperparameter Optimisation

Hyperparameters were tuned via Weights & Biases Bayesian sweeps, maximising validation AUC $\uparrow$ on fold 0. The search space included learning rate, weight decay, dropout, hidden dimensions, number of selected genes, label smoothing, batch size, weighted loss, weighted sampling, and early stopping patience.

3.2 Evaluation Protocol

The best configuration was evaluated using stratified 10-fold cross-validation with patient-level splitting. Each patient appears in exactly one test fold. Aggregating predictions across the ten folds therefore yields held-out predictions for the full RNA cohort without leakage across train and test partitions.

4 Results

4.1 Per-Fold Validation and Test Metrics

Each fold used 837 training, 104 validation, and 104 test samples with 10,000 selected genes. Table 3 shows the per-fold validation and test metrics.

Table 3: Per-fold RNA-seq validation and test results.

Fold	Val AUC $\uparrow$	Val Acc. $\uparrow$	Test AUC $\uparrow$	Test Acc. $\uparrow$
0	0.9982	0.9519	0.9711	0.9038
1	0.9911	0.9615	0.9873	0.9135
2	0.9922	0.9231	0.9828	0.8942
3	0.9877	0.9038	0.9924	0.9135
4	0.9888	0.8942	0.9839	0.8750
5	0.9784	0.8750	0.9751	0.8846
6	0.9918	0.8942	0.9869	0.8942
7	0.9947	0.9038	0.9858	0.9135
8	0.9897	0.9231	0.9757	0.8654
9	0.9841	0.8750	0.9767	0.8558
Mean $\pm$ SD	0.9897 $\pm$ 0.0055	0.9106 $\pm$ 0.0294	0.9818 $\pm$ 0.0068	0.8913 $\pm$ 0.0208

4.2 Aggregated Predictions

Since each RNA sample appears in exactly one test fold, we aggregate all test predictions across the 10 folds to obtain held-out predictions for the full cohort. The aggregate held-out test performance is:

- Macro AUC $\uparrow$: 0.9821
- Overall accuracy $\uparrow$: 0.8913
- Weighted F1-score $\uparrow$: 0.8930

4.3 Per-Fold Test AUC $\uparrow$

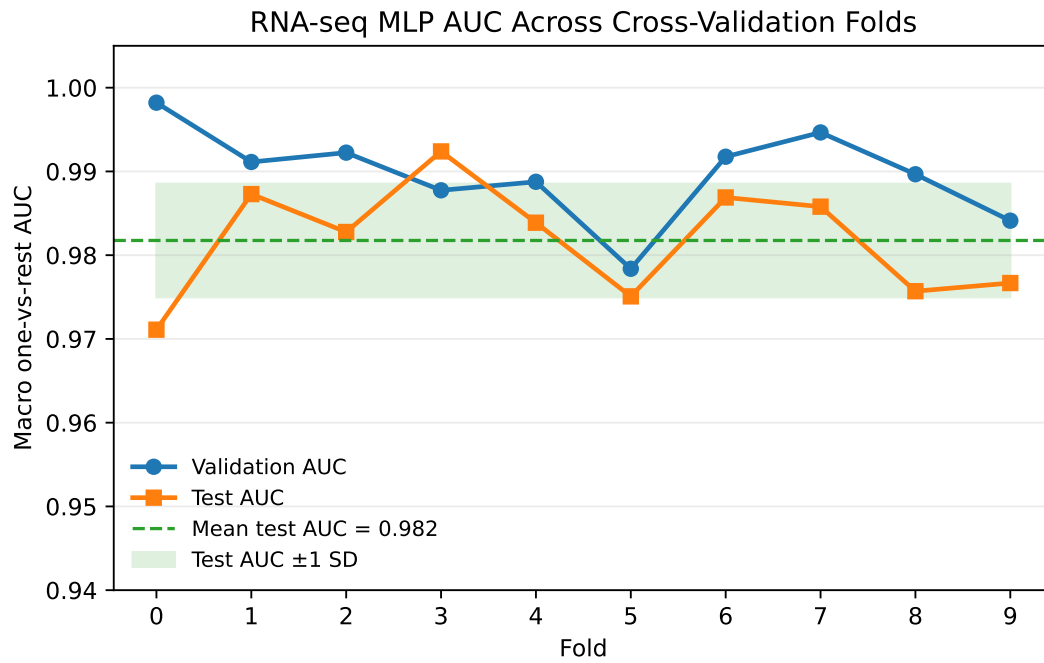


Figure 1: Validation and test AUC $\uparrow$ across folds. The dashed line denotes mean test AUC $\uparrow$ and the shaded region denotes one standard deviation.

4.4 Confusion Matrix

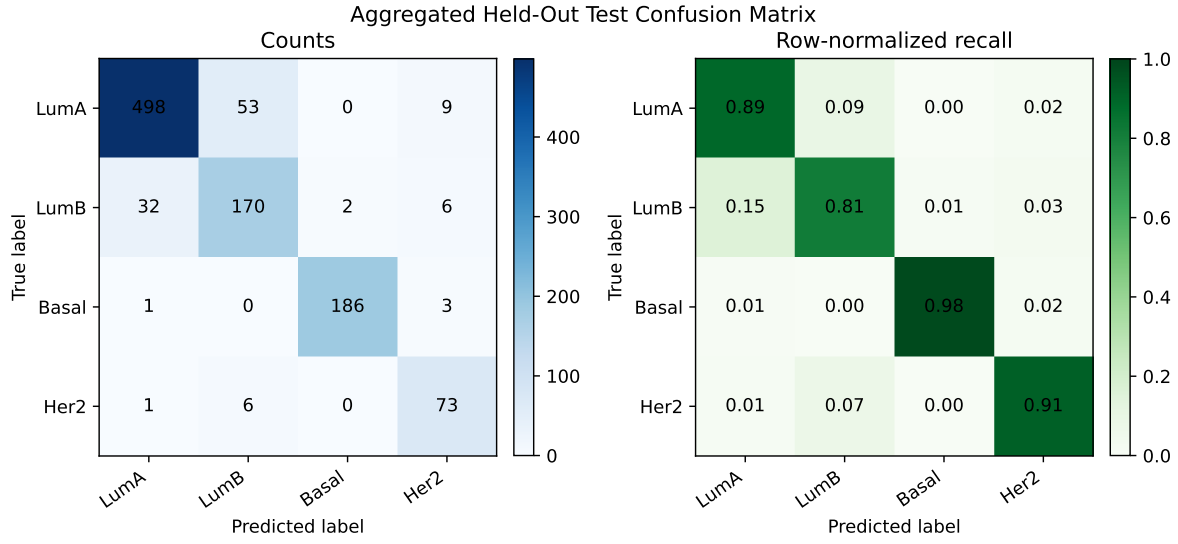


Figure 2: Confusion matrix for aggregated 10-fold held-out test predictions. Left: raw counts. Right: row-normalised recall $\uparrow$.

4.5 Classification Report

Table 4: Per-class test metrics from aggregated held-out predictions.

Class	AUC $\uparrow$	Precision $\uparrow$	Recall $\uparrow$	F1 $\uparrow$	Support
LumA	0.9782	0.936	0.889	0.912	560
LumB	0.9621	0.742	0.810	0.774	210
Basal	0.9991	0.989	0.979	0.984	190
Her2	0.9889	0.802	0.912	0.854	80

4.6 ROC Curves

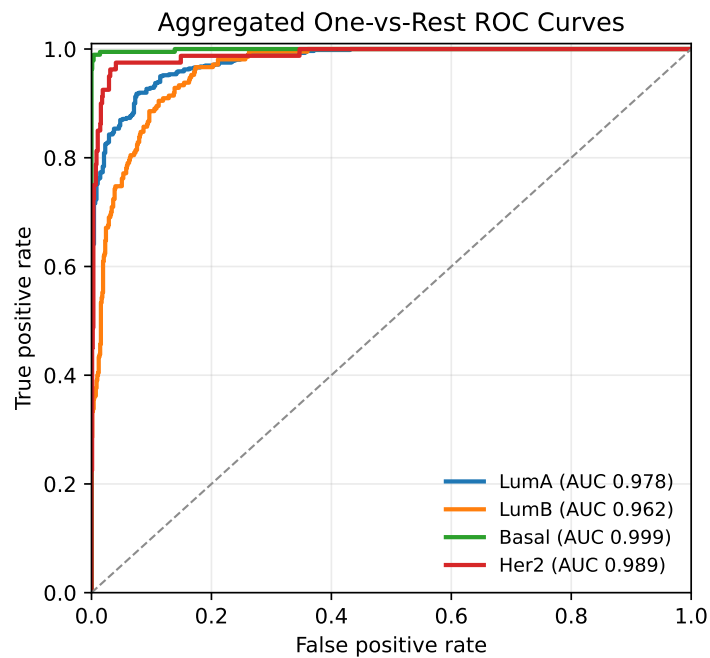


Figure 3: One-vs-rest ROC curves computed from aggregated held-out test predictions.

4.7 Per-Class AUC $\uparrow$ Across Folds

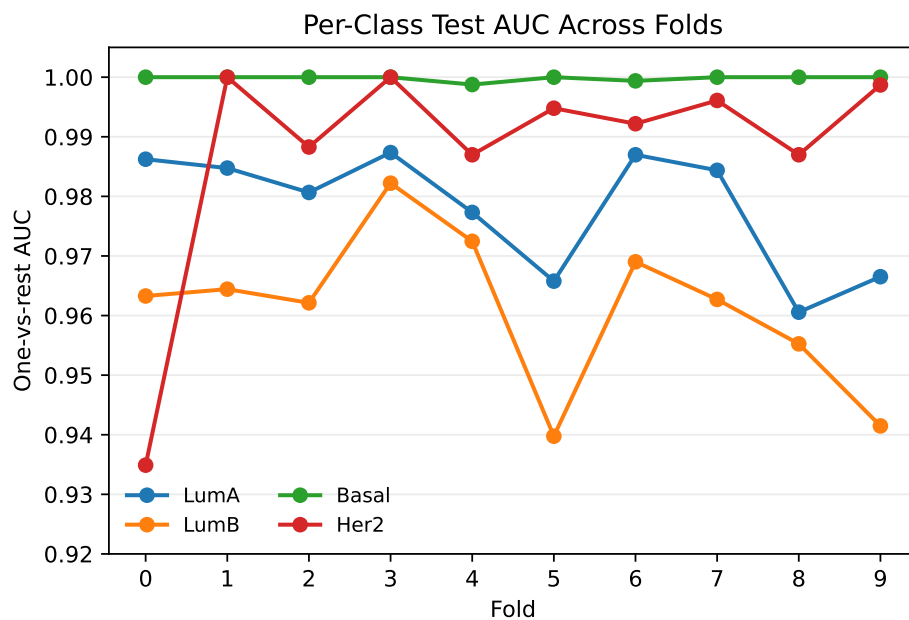


Figure 4: Per-class test AUC $\uparrow$ across the 10 folds.

5 Comparison to the WSI-only Branch

The RNA-only branch is compared with the existing WSI-only CLAM-MB + UNI2-h branch. The WSI branch remains the primary thesis focus because it predicts molecular subtype from histology. RNA-seq is used here as a strong molecular reference and future fusion modality. For the visual comparison, we use macro AUC, balanced accuracy, and macro F1 because the task is four-class and class-imbalanced. WSI balanced accuracy and macro F1 are computed from the aggregated per-class recall and F1 values.

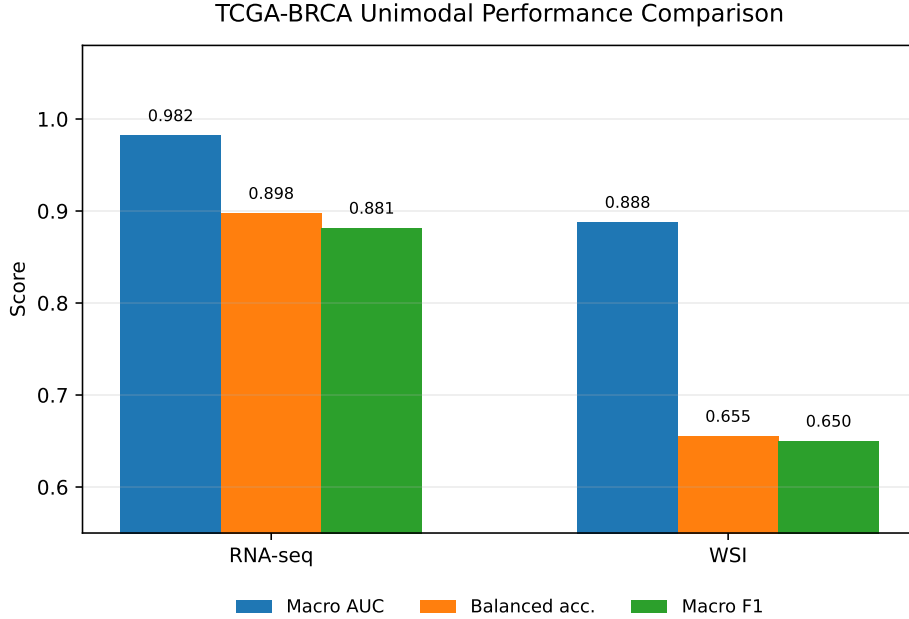


Figure 5: Comparison of RNA-seq and WSI unimodal branches using class-sensitive metrics. Higher values are better.

6 WSI-RNA Fusion Evaluation

The fusion experiments use the 986 slides with both WSI embeddings and matched RNA-seq features. Evaluation follows the same 10-fold patient-level protocol. Table 5 reports aggregate held-out test metrics after concatenating predictions from all folds. The matched RNA row is the unimodal RNA baseline retrained and evaluated on the same slide-level cohort and splits as the fusion models.

Table 5: Aggregate held-out test results for WSI-RNA fusion.

Method	n	AUC $\uparrow$	Acc. $\uparrow$	Bal. Acc. $\uparrow$	Macro F1 $\uparrow$	Weighted F1 $\uparrow$
Matched RNA only	986	0.9785	0.8783	0.8809	0.8644	0.8799
WSI + RNA concat	986	0.9765	0.8884	0.8745	0.8653	0.8896
WSI + RNA gated	986	0.9744	0.8874	0.8724	0.8667	0.8885
Late probability fusion	986	0.9789	0.8803	0.8756	0.8657	0.8812
WSI + RNA residual	986	0.9787	0.8803	0.8535	0.8575	0.8797

Across folds, concat fusion achieved test AUC 0.9783 ± 0.0062 and accuracy 0.8890 ± 0.0276 . Gated fusion achieved test AUC 0.9751 ± 0.0065 and accuracy 0.8879 ± 0.0282 . Late probability fusion achieved aggregate AUC 0.9789 and accuracy 0.8803. Residual fusion achieved aggregate AUC 0.9787 and accuracy 0.8803. The fusion methods therefore recover most of the RNA signal, but they do not clearly improve over the matched RNA baseline across all metrics.

For the gated model, the aggregate mean WSI gate is 0.4709 and the mean RNA gate is 0.5291. The gate is similar across subtypes, with mean WSI gates between 0.4661 for LumA and 0.4793 for LumB. Correct predictions have mean WSI gate 0.4703, while incorrect predictions have 0.4757, so the learned gate does not behave as a strong confidence switch in this run.

6.1 Residual Fusion

A 25-run Bayesian sweep tested RNA-preserving residual fusion on fold 0. The RNA and WSI branches were frozen, and only a small residual head was trained. The validation-selected configuration used residual scale 0.10, fusion hidden dimension 16, dropout 0.25, learning rate 0.000165, and weight decay 0.00000691. This configuration was then evaluated with the same 10-fold protocol.

Table 6: Fold-level 10-fold comparison for residual fusion.

Method	Test AUC $\uparrow$	Test Acc. $\uparrow$
Matched RNA only	0.9806 ± 0.0075	0.8787 ± 0.0285
WSI + RNA concat	0.9783 ± 0.0062	0.8890 ± 0.0276
WSI + RNA gated	0.9751 ± 0.0065	0.8879 ± 0.0282
Late probability fusion	0.9806 ± 0.0067	0.8806 ± 0.0268
WSI + RNA residual	0.9799 ± 0.0090	0.8804 ± 0.0286

Residual fusion slightly improved mean accuracy over the matched RNA baseline, but did not improve mean test AUC. On aggregated predictions it reached 0.8535 balanced accuracy, 0.8575 macro F1, and 0.8797 weighted F1, all below the strongest matched RNA or concat metrics. The result supports the same conclusion as concat, gated, and late probability fusion: WSI contributes some signal, but the current fusion heads do not consistently improve over the RNA branch.

7 Comparison to SOTA

Hezil et al. proposed *Selective multimodal deep learning for reliable breast cancer subtype classification from histopathology and genomic data* [1], a TCGA-BRCA PAM50 pipeline combining CTransPath WSI features, RNA-seq features, multimodal fusion, and confidence-aware routing. Tables 7 and 8 separate the single-modality and multimodal comparisons.

Table 7: Single-modality comparison on TCGA-BRCA PAM50 subtype classification. Literature metrics are from Hezil et al. [2]. Best values are bolded within comparable modality settings.

Method	Setting	Cohort	Accuracy $\uparrow$	Macro F1 $\uparrow$	Weighted F1 $\uparrow$
<i>Reference methods</i>					
Hezil et al. [2]	RNA-seq only	1022 patients	0.9100	0.90	0.91
Hezil et al. [2]	WSI only, CTransPath	1022 patients	0.6200	0.57	0.64
<i>Our methods</i>					
RNA-seq	RNA-seq MLP	1045 samples	0.8913 ± 0.0208	0.8815 ± 0.0292	0.8929 ± 0.0198
Vision Branch	UNI2-h + CLAM-MB	986 slides	0.7181	0.6500	0.7179

Table 8: Multimodal comparison on TCGA-BRCA PAM50 subtype classification.

Method	Setting	Cohort	Accuracy $\uparrow$	Macro F1 $\uparrow$	Weighted F1 $\uparrow$
Hezil et al. [2]	Multimodal concatenation	1022 patients	0.9350	0.92	0.93
Hezil et al. [2]	Multimodal gated fusion	1022 patients	0.9420	0.93	0.94
Hezil et al. [2]	Multimodal cross-attention	1022 patients	0.9480	0.93	0.95
Hezil et al. [2]	Routing-based fusion	1022 patients	0.9505	0.93	0.95
Hezil et al. [1]	Routing-based fusion	924 patients	0.9493	—	—
<i>Our methods</i>					
RNA-seq, matched WSI cohort	RNA-seq MLP	986 slides	0.8783	0.8644	0.8799
WSI + RNA concat	UNI2-h CLAM-MB + RNA MLP	986 slides	0.8884	0.8653	0.8896
WSI + RNA gated	UNI2-h CLAM-MB + RNA MLP	986 slides	0.8874	0.8667	0.8885
Late probability fusion	RNA MLP + UNI2-h CLAM-MB	986 slides	0.8803	0.8657	0.8812
WSI + RNA residual	Frozen RNA + residual WSI head	986 slides	0.8803	0.8575	0.8797

RNA-seq is the stronger single-modality baseline for PAM50, while multimodal fusion gives the highest reported accuracies in Hezil et al. In our experiments, concat, gated, late probability, and residual fusion recover almost all of the RNA-only performance on the matched WSI-RNA cohort, but they do not provide a consistent improvement over the RNA-only branch. These values are not a direct ranking because cohort size, splits, feature extractors, and reporting protocols differ.

This result is consistent with prior work showing that RNA and other omics modalities contain strong subtype signal. Lin et al. [4] proposed DeepMO for breast cancer subtype classification from mRNA, methylation, and copy-number alteration data, reporting that mRNA and methylation are highly informative for subtype classification. Ektafaie et al. [3] studied histology-only and multiomics-histopathology breast cancer classification and reported four-class PAM50 prediction from H&E images, supporting the histology-first framing. Recent multimodal studies further support attention- or routing-based fusion of image and molecular or clinical modalities [5], although the precise datasets, subtype groupings, and validation protocols differ.

8 Summary and Interpretation

The RNA-seq branch achieved strong 10-fold performance on TCGA-BRCA four-class PAM50 classification:

- Validation macro AUC $\uparrow$: 0.9897 ± 0.0055
- Validation accuracy $\uparrow$: 0.9106 ± 0.0294
- Test macro AUC $\uparrow$: 0.9818 ± 0.0068
- Test accuracy $\uparrow$: 0.8913 ± 0.0208
- Test balanced accuracy $\uparrow$: 0.8976 ± 0.0284

This performance is consistent with the fact that PAM50 is defined from gene expression. In our approach, the RNA-only model serves as a molecular reference branch and as the basis for fusion with WSI features. The histology branch addresses the main image-based task: predicting molecular subtype from H&E whole-slide images, with the RNA branch providing a complementary modality for multimodal learning.

On the matched 986-slide cohort, RNA-only achieved 0.8783 accuracy and 0.9785 macro AUC. Concat fusion improved accuracy to 0.8884 but reduced macro AUC to 0.9765. Gated fusion reached 0.8874 accuracy and 0.9744 macro AUC. Late probability fusion gave the highest aggregate macro AUC among fusion methods at 0.9789, but only a small accuracy gain over matched RNA. The validation-selected residual model reached 0.9787 aggregate macro AUC, 0.8803 accuracy, and 0.8575 macro F1. Residual fusion therefore also remains close to RNA, but does not consistently improve on it. This suggests that the next useful step is not a larger fusion head by itself, but confidence-aware use of histology signal where RNA is uncertain.

References

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